

Assignment: Sentiment Analysis with VADER

PolitiPulse NLP Workshop — Week 3 Follow-Up Assignment

Course Module:	NLP for PolitiPulse (Week 3)
Topics Covered:	Text preprocessing, VADER sentiment scoring, approval-rating pipeline
Tools Required:	Python 3, vaderSentiment library
Total Marks:	20

Instructions

- Answer all three questions. Each question is code-based — write working Python code for every part and include the printed output.
- Install the library first if you haven't: `pip install vaderSentiment`
- Use the standard import shown in the workshop: `from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer`
- Do not use stemming, lemmatization, POS tagging, or any ML/deep-learning model — this assignment stays within the scope covered in the workshop (VADER only).

Question 1 — Preprocessing Function + VADER Scoring (6 marks)

Below are five raw public comments collected by PolitiPulse's Layer 1 (Data Acquisition):

- "The New Roads Policy is AMAZING!!!"
- "i dont think the new tax rule helps anyone."
- "The committee met on Tuesday to review the budget."
- "Honestly, this healthcare reform is a disaster."
- "Farmers finally got the support they deserve, great move!!!"

(a) Write a function `preprocess(text)` that applies the three preprocessing steps taught in the workshop — lowercasing, punctuation removal, and tokenization — and returns a list of tokens. Run it on all five comments and print the token list for each. (3 marks)

(b) Write Python code that runs VADER's `polarity_scores()` on all five (original, non-preprocessed) comments and prints the compound score for each. (3 marks)

Question 2 — Build a Mini Approval-Rating Pipeline (10 marks)

The workshop described PolitiPulse's pipeline as: *Public Comment* → *VADER* → *Sentiment Score* → *Average Score* → *Approval Rating* → *Dashboard*. In this question you will build a simplified version of that pipeline.

(a) Write a function `get_compound_scores(comments)` that takes a list of comment strings and returns a list of their compound scores using VADER. (3 marks)

(b) Write a function `average_score(scores)` that returns the mean of a list of compound scores. (2 marks)

(c) Write a function `approval_rating(avg_score)` that maps the average compound score to one of five labels using these thresholds, then returns the label:

Average Compound Score	Approval Rating Label
$\text{avg} \geq 0.5$	Strongly Approve
$0.05 \leq \text{avg} < 0.5$	Approve
$-0.05 < \text{avg} < 0.05$	Neutral
$-0.5 < \text{avg} \leq -0.05$	Disapprove
$\text{avg} \leq -0.5$	Strongly Disapprove

(3 marks)

(d) Using the five comments from Question 1, run your full pipeline (compute compound scores → average → approval rating) and print the final approval rating label. (2 marks)

Question 3 — Per-Comment Sentiment Classifier (4 marks)

The workshop's Sentiment Analysis Fundamentals slide labels text as Positive, Negative, or Neutral based on its score. Write a function `classify_sentiment(text)` that:

- runs VADER on text to get its compound score,
- returns "Positive" if compound ≥ 0.05 ,
- returns "Negative" if compound ≤ -0.05 ,
- returns "Neutral" otherwise.

Apply `classify_sentiment()` to each of the five comments from Question 1 and print the comment alongside its label.

End of Assignment